

# ARTHUR HENRIQUE LOPES FEITOSA

## Backend AI Engineer | Java (Spring Boot) & Python (FastAPI) | LLMs & Microservices

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### PROFESSIONAL SUMMARY

Information Systems student and Backend AI Engineer with hands-on experience developing high-throughput asynchronous REST APIs, event-driven worker queues, and autonomous LLM diagnostic loops. Strong technical foundation in Java (Spring Boot), Python (FastAPI), PostgreSQL, Docker, and Anthropic Claude APIs. Proven track record in achieving 100% automated resolution pass rates, reducing software defects, and integrating OCR document analysis pipelines.

### TECHNICAL SKILLS

**Backend & Microservices:** Java, Spring Boot, Python, FastAPI, RESTful APIs, Microservices Architecture, Spring Security, JWT Authentication, Celery, Inngest.

**Databases & Cloud:** PostgreSQL, Redis Cache, Supabase, SQL, Docker, AWS (EC2, S3 fundamentals), Git, GitHub Actions, CI/CD.

**AI & Agent Systems:** Anthropic Claude API, Model Context Protocol (MCP), Autonomous SRE Repair Agents, Prompt Engineering, OCR LLM Extraction.

**Testing & Methodologies:** JUnit, Mockito, Pytest, Swagger/OpenAPI, Clean Code, SOLID Principles, Agile Scrum.

**Languages:** Portuguese (Native), English (C1 Advanced — Professional Technical Interpreter).

### WORK EXPERIENCE

#### Backend AI Engineering Intern

July 2026 – Present

*FlyRank AI Internship*

*Remote*

- Architected and deployed the 10x Task Manager Platform and Autonomous SRE Agent using FastAPI and PostgreSQL with sub-50ms P95 latency.
- Engineered self-healing LLM triage loop diagnosing exceptions and applying automated code patches with 100% resolution pass rate.
- Implemented asynchronous event processing queues using Celery and Redis for PDF report generation and automated multi-modal tagging.

#### Software Development Intern

June 2025 – June 2026

*Midas Systems Development*

*Remote*

- Developed an intelligent conversational chatbot using Spring Boot, PostgreSQL, and Flutter/Dart, improving user interaction by 20%.
- Integrated LLM document processing pipeline via OCR, increasing automated extraction accuracy on unstructured records by 15%.
- Wrote comprehensive unit and integration testing suites with JUnit and Mockito, reducing post-release defect rate by 30%.

#### English Interpreter

January 2024 – December 2024

*FIRA RoboWorld Cup 2024*

*Brazil*

- Delivered real-time technical interpretation during international robotics competitions, facilitating mission-critical communication for engineering teams.

### PROJECTS

**Task Manager API & Autonomous SRE Agent — Lead Developer**  
*Python, FastAPI, PostgreSQL, Supabase JWT, Celery, Redis, React Flow, Claude Code*

2026  
*Open Source*

- Built asynchronous task orchestration platform with interactive React Flow DAG visualization and automated Claude-driven fault triage.
- Authored full unit and integration test coverage with Pytest and published complete OpenAPI documentation on GitHub.

**SIGAMA Vision: AI Agricultural Compliance Platform — Developer**  
*React 18, TypeScript, Tailwind CSS, Python OCR, LLMs*

2025  
*AGED / FAPEMA Grant*

- Developed document verification workflow for agricultural transit guides (GTAs), generating automated compliance scores and fraud detection flags.

**LLMX: Terminal AI Assistant for Linux & DistroWiki — Developer**  
*TypeScript, Node.js, React Ink, React, Vite, Tailwind CSS*

2024 – 2025  
*Open Source*

- Constructed LLMX for natural language command execution on Linux and DistroWiki for side-by-side Linux distribution benchmarking.

EDUCATION

**Bachelor of Science in Information Systems**  
*Federal Institute of Maranhão (IFMA)*

March 2024 – March 2028  
*São Luís, MA, Brazil*

CERTIFICATIONS

- **Anthropic Academy:** 20 Certified Credentials in Claude Architecture, Model Context Protocol (MCP) & AI Agent Systems (2026).
- **Digital Innovation One (DIO):** 14 Certifications in Java, Spring Boot, Microservices, REST APIs, SOLID & Clean Code Architecture.
- **AWS:** Introduction to AWS Cloud Computing & EC2 Architecture Fundamentals.